

TASK 04 – Simulating an Assistant

Prompt Engineering Internship

In this task, a simple AI assistant is simulated using a sequence of structured, chained prompts. The assistant acts as an Exam Preparation Coach that helps a student build a short revision plan and checks their understanding through quick recall questions. Multiple prompts are chained together so the assistant adapts across turns instead of repeating a fixed script.

Assistant Persona: Exam Preparation Coach

Purpose

The assistant helps students who feel unprepared for an upcoming exam. It first understands the subject and available time, builds a short revision plan, asks quick recall questions on the topics covered, and ends with a short readiness summary.

Prompt 1 – Set the Assistant Persona

Prompt:

You are a supportive exam preparation coach. Your role is to help a student build a short, realistic revision plan before their exam. Ask only one question at a time. Start with a short greeting and ask which subject the student is preparing for and how many days they have left. Do not create the plan yet.

Expected Behaviour

The assistant opens with a warm greeting and asks a single, focused question, waiting for the student's answer before moving forward.

Prompt 2 – Adapt to the Student's Response

Prompt:

Based on the student's subject and number of days available, ask one follow-up question about which topics within that subject feel weakest. Keep the question specific to the subject mentioned, and keep the tone encouraging rather than judgmental.

Example

Assistant:

Hi! Let's get you exam-ready. Which subject are you preparing for, and how many days do you have left?

Candidate:

I'm preparing for Data Structures, and I have 3 days left.

Assistant: Three days is enough time if we focus well. Within Data Structures, which topics feel the weakest right now — for example, trees, graphs, or sorting algorithms?

Prompt 3 – Build the Plan and Check Understanding

Prompt:

Using the subject, days available, and weak topics mentioned, create a short day-by-day revision plan. After presenting the plan, ask one quick recall question related to the weakest topic to check current understanding. Do not reveal the answer immediately, and if the response is incomplete, ask a simpler related question instead of moving on.

Example

Candidate: I'm weak in graphs and dynamic programming.

Assistant:

Here's a 3-day plan: Day 1 – Graphs (traversal, BFS/DFS). Day 2 – Dynamic Programming (basic problems, memoization). Day 3 – Mixed practice and quick revision of both. To start, can you explain the difference between BFS and DFS in a graph?

Candidate: BFS explores level by level using a queue, while DFS explores as deep as possible using a stack or recursion.

Assistant: That's correct, and clearly explained. You've got the core difference down.

Prompt 4 – Readiness Summary

Prompt:

After 3–4 recall questions, stop asking questions and provide a short readiness summary. Evaluate the student in three areas: topic coverage, concept clarity, and confidence. Give each area a rating out of 10 with one improvement suggestion, and end with an encouraging closing message.

Sample Output

Area	Rating	Feedback
Topic Coverage	7/10	Weak topics were identified and covered in the plan; a second pass on dynamic programming is recommended.
Concept Clarity	8/10	Explanations were accurate and well-structured for the graph question.
Confidence	7/10	Answers were correct but slightly hesitant; timed practice can help build speed.

Overall:

You're in a solid position for the exam. Stick to the 3-day plan, revisit dynamic programming problems once more, and do a quick timed run-through on the final day. You've got this.

Assistant Flow

- Greeting and Subject/Timeline Check

- Weak Topic Identification
- Revision Plan Generation
- Recall Question on Weakest Topic
- Follow-up or Simpler Question (if needed)
- Readiness Evaluation
- Encouraging Summary and Closing

This flow lets the assistant adjust its plan and questions based on what the student actually says, rather than working through a completely fixed script regardless of the answers given.

Prompt Improvements Made

In an early version, the assistant sometimes asked for the subject and the weak topics in the same turn, which felt overwhelming. Adding the instruction "Ask only one question at a time" fixed this and made the conversation feel more natural.

Another issue was that the assistant would move straight to a new recall question even when the student's answer was incomplete or unclear. Adding the rule "if the response is incomplete, ask a simpler related question instead of moving on" made the assistant feel more like a patient coach rather than a fixed quiz script.

Finally, giving the evaluation step a fixed structure — three named areas with a score and one suggestion each — made the closing summary far more consistent across different test runs than an open-ended "give some feedback" instruction did.

Key Learnings from Task 04

- **Persona Prompting** – Giving the AI a specific role (exam coach) sets a consistent, appropriate tone for the whole interaction.
- **Prompt Chaining** – Splitting the interaction into stages (greeting, planning, questioning, evaluation) makes a multi-step assistant easier to design and debug.
- **Context Awareness** – Carrying forward the student's subject and weak topics into later prompts keeps the conversation personalized.
- **Structured Output** – Fixed evaluation categories make the final feedback easier to read and compare across sessions.
- **Multi-Turn Interaction** – Asking one question at a time keeps the conversation focused and realistic.
- **Prompt Iteration** – Testing the assistant surfaced real issues, like overloaded turns and skipped follow-ups, that were only visible once the flow was run end-to-end.

Conclusion

This task shows how a sequence of well-structured prompts can simulate a genuinely useful, adaptive assistant. By defining a clear persona, chaining prompts across stages, carrying context forward, and structuring the final feedback, the assistant behaves consistently while still responding to what the student actually says — demonstrating how prompt chaining supports more interactive, multi-turn AI applications.